



Clinical Practice Guidelines

Nebulisation

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	All Staff

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Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to nebulisation. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline may:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☒
Other:	☒		

1. Principles

All prescriptions written within Sir Charles Gairdner Osborne Park Health Care Group are to meet the minimum requirements set under the Medicines and Poisons Act 2014 and Medicines and Poisons Regulations 2016, [SCGOPHCG Prescribing Policy and Procedure](#) in addition to best practice requirements.

Nurses endorsed as a nurse practitioner with the Nurses and Midwives Board of Australia working in a designated area have the legislative authority to prescribe scheduled medicines, within professional and personal scope of practice.

A Registered Nurse/ medication competent Enrolled Nurse may only initiate Schedule 2 non-prescription and non-scheduled medications *that are approved and listed* within the 'Endorsed List of Nurse Initiated Medication' nursing practice guideline without medical officer or nurse practitioner written authorisation.

Observe the 5 moments for hand hygiene. Use of appropriate personal protective equipment (PPE) – this includes gloves, plastic aprons, long sleeved fluid repellent gowns, surgical mask or respirator and protective eyewear when there is a likelihood of contact with blood, other body fluids, secretions and excretions (except sweat), non-intact skin and mucous membranes.

2. General Management

2.1 Medical Requirements

The medical officer is to prescribe drugs for nebulisation on the WA Hospital Inpatient Medication Chart.

The preferred method of delivery of short acting beta 2 agonist (e.g. Salbutamol™ Ventolin®) and anticholinergic (e.g. Ipratropium Atrovent®) bronchodilators is using a pressurised metered-dose inhaler (pMDI) via a spacer.¹

2.2 Terminology

Air Compressor Unit	A device that compresses room air and forces it through a liquid medication to turn it into a mist of fine droplets which is then inhaled via a nebuliser
Inhaler	Hand-held device which delivers medications to lungs via dry powder aerosol, powder or mist when inhaled using correct technique
Nebuliser	A device which converts a liquid into aerosol droplets suitable for patient inhalation and delivery of solution to the airway ²
pMDI	Pressurised metered dose inhaler
Spacer	A spacer is a tube-shaped device that acts as a holding chamber between the mouth and cannister of medication. It is used to increase the ease of giving inhaled medications from pMDIs. (Image 1)



Figure 1. Spacer with pMDI attached

2.3 Safety Information

Nebulisation can stimulate coughing and promote the generation of respiratory aerosols which can increase the risk of transmission of infections (e.g. COVID-19, Influenza like illnesses) to staff patients and visitors.

For patients with suspected or confirmed acute respiratory infections (ARIs) such as Influenza Like Illnesses (ILI) or COVID-19 administration of medication via nebulisation is not recommended. Where possible pMDI's and spacer should be used unless nebuliser is authorised by the medical consultant. Patients with ARIs require single room with transmission-based precautions. Refer to the Infection Prevention and Management Manual for further information.

Patients being nursed with transmission-based precaution who have a multi-resistant organism isolated in respiratory specimens require the door to be shut during nebulisation therapy and to remain shut for 30 minutes⁶ after completion. Appropriate personal protective equipment (PPE) e.g., mask/respirator and protective eyewear may be required to be worn.

When handling a nebuliser and dispensing medication, an aseptic non-touch technique must be followed. Single use nebulisers should be used wherever possible to avoid the need to draw solutions into a syringe prior to administration.³

Medication for inhalation should be made available in single-use nebulisers to avoid measurement in a syringe, however if measurement via a syringe is required, ensure the syringe is non-luer lock and is clearly labelled "For Inhalations Use Only"(figure2.)^{3,4} A purple, enteral syringe is the preferred device to be used.

For Inhalation Use Only				
Patient				
ID	DOB			
Medicine/s	Amount (units)	÷	Volume (mL)	= Conc (units/mL)
.....				
.....				
.....				
Diluent				
Date	Prepared by			
Time	Checked by			

Figure 2: Inhalation medicine label

Nebulised antimicrobials must be administered via a high efficiency filter system and in a single room⁴. Don a N95 mask and eye wear. Filters must be discarded after each use.

The door to the patient's room should be closed during the therapy and remain closed for 30 minutes after completion of the therapy to prevent occupational exposure to staff.

Monitor patients receiving nebulised antibiotics for the first time for bronchospasm.¹

Masks (with straps) are recommended for use in:

- Acutely ill patients, or
- When patient co-ordination is difficult.

Mouthpieces are recommended with nebulised:²

- Steroids – to prevent deposition on the face.
- Antimicrobials – with a high efficiency filter, to prevent exhalation of antibiotics into the air.
- Anticholinergics – such as Ipratropium Bromide (Atrovent™), as they may exacerbate glaucoma.

The mouth should be rinsed out with tap water following the administration of nebulised steroids to prevent local effects such as oral thrush.^{1,6}

Some nebulised medicines such Iloprost™, Pentamidine and nebulised antibiotics may pose an occupational hazard for staff and have specific guidelines. Consult the Respiratory CNS or a pharmacist with any queries or drug specific guidelines.

Occupational exposure can result in bronchospasm or asthma type symptoms.¹

2.4 Indications

- Nebulisation treatment may be considered when a patient requires very high doses of a bronchodilator, inhaled medication is not able to be given via other means or if the patient is unable to use other inhaler devices effectively due to age, inspiratory effort, cognition, dexterity and neuromuscular conditions^{1,6,7}.
- Patient factors such as breathing pattern, artificial airway, airway obstruction and nose vs mouth breathing should also be considered when administering nebulisers and choosing the appropriate interface.⁸
- Nebulised treatment provides patients with an alternative administration route that avoids the need for inspiratory flow, manual dexterity, or complex hand-breath coordination⁸

2.5 Nursing Management

Nebulisation can be administered via:

- Air Compressor Unit – Figure 3 or,
- Piped medical air/oxygen at 6-8L/minute^{1,2}- Figure 4
- If the patient is on oxygen via nasal prongs, give medication by nebuliser (compressor) unit by applying face mask over the top. Continue nasal oxygen as prescribed.



Figure 3: Air Compressor Unit



Figure 3: Piped medical oxygen/medical air flow meters

- If the patient needs continuous oxygen therapy over 6L/minute, administer the nebuliser with oxygen – do not exceed 8-10L/min and avoid over-oxygenation due to increased risk of hypercapnoea.
- In patients experiencing acute asthma with acute-severe or life-threatening features, the nebulised route (oxygen driven) is recommended.¹
- Stay with patient if they are unable to manage nebuliser independently.

2.6 Procedural Information

Requirements

- Airflow meter/Oxygen flow meter or Air Compressor Unit
- Nebuliser reservoir and T piece with mouthpiece, facemask or tracheostomy interface (Figure 5)
- Nebuliser tubing
- Patient's medication chart
- Prescribed drug and diluent if required.
- High efficiency filters (if required)
- Sterile Water (for rinsing nebuliser)
- Purple enteral syringe
- PPE as for Standard precautions, including facial protection.



Figure 4. Nebuliser in order 1) T-piece with mouthpiece, 2) Face mask, 3) Tracheostomy / laryngectomy interface

Procedure ^{9,10}

1. Perform hand hygiene.
2. Check medication chart as to prescribed medication, dose, and diluent.
3. Explain the procedure to the patient and gain consent.
4. Assist the patient into a comfortable position, preferably sitting upright or raise head of bed to at least 30° if medically appropriate.
5. Perform hand hygiene and don PPE.
6. Attach one end of the nebuliser tubing to the air compressor unit or the air/oxygen flowmeter nipple. Attach the other end to the base of the nebuliser bowl. Undo the top of nebuliser and add solution to be nebulised into the nebuliser reservoir. If nebuliser is not available, draw-up medication with an enteral syringe and compatible non-luer straw (not a luer-lock needle and syringe) using a non-touch technique and apply a medication route label "For Inhalation Use Only".
7. Replace top of the nebuliser and attach interface (mask, T-piece, tracheostomy mask/mount). Turn on the air compressor unit or turn air or oxygen flow meter to 6-8L/min, and check that mist is issuing from the nebuliser.
8. Apply nebuliser to the patient and instruct the patient to take normal steady breaths through the mouth, and to keep the nebuliser upright.
 - a. If a T-piece with mouthpiece is used, instruct the patient to maintain a tight seal with their teeth and lips around the mouthpiece and breathe through the mouth.
 - b. If a facemask is used, adjust the metal support over the bridge of the nose to ensure a good mask fit.
 - c. In patients with a tracheostomy a mask can be placed over the tracheostomy opening or the nebuliser chamber can be attached to the tracheostomy tube using a t-piece. The T—piece approach is preferred as more medication is directed into the tracheostomy tube⁵
9. If it is the first dose, observe the patient during the procedure for any change in condition.
10. Nebuliser should be complete within 5-10 minutes. There will be a small amount of fluid left in the chamber

11. On completion, remove nebuliser and turn off Medical Air/ Medical Oxygen/ Air Compressor Unit.
12. Document in the medical records.
13. Immediately clean nebulizer.

Cleaning and Maintenance of Nebulisers¹¹

Nebulisers should be cleaned and air-dried in a clean environment **immediately after each use.**

Discard the nebuliser if:

- It is visibly soiled or damaged.
- Suspected contamination has occurred (e.g. dropped on the floor)

Nebuliser tubing should not be washed. Replace if visibly soiled.

Procedure

1. Perform Hand Hygiene and don gloves.
2. Disassemble the nebuliser:
 - a. Remove the mask or mouthpiece
 - b. Separate the nebuliser chamber.
 - c. Inspect for visible damage or residue/soiling, **discard if present.**
3. Rinse all parts (except the tubing) with sterile water to remove any residual medication.
4. Allow to air dry completely:
 - a. Place parts on a clean, dry surface in a designated area.
 - b. Do not towel dry.
5. Reassemble when fully dry.
6. Discard the nebuliser:
 - a. After 7 days, or
 - b. On completion of nebuliser treatment, or
 - c. On patient discharge - **whichever occurs first.**

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010
- Occupational Health, Safety and Welfare Act 1984.
- Poisons Act 1964, Poisons Regulations 1965 and Poisons Amendment Regulations 2010

Policy:

- WA Health Mandatory Policy 0131/20 High Risk Medication Policy.
- NMHS Clinical Documentation Policy
- SCGOPHCG Prescribing Policy and Procedure
- SCGOPHCG Medication Administration Policy
- [SCGOPHCG High Risk Medication Policy](#)
- [SCGOPHCG Medicine Management](#) Policy
- SCGOPHCG Infection Prevention and Control Manual -Aseptic Technique Policy.
- SCGOPHCG Infection Prevention and Control Manual: Influenza-like Illness Policy
- SCGOPHCG Infection Prevention and Management Manual COVID-19 v 14.2
- SCGOPHCG Infection Prevention and Control Policy Respiratory Viruses (Non-Endemic)
- SCGH Infection Prevention and Control Manual: Cleaning of Patient Equipment
- OPH Infection Prevention and Control Manual: Patient care equipment and reusable medical devices: Reprocessing standards
- SCGOPHCG Patient Identification Policy

- SCGOPHCG Procedure Matching Policy
- [National Standard for User applied Labelling of Injectable Medicines, Fluid and Lines](#)
- Western Australian Health Operational Directive OD0441/13, WA. Safe administration of oral, enteral or nebulizer solutions (use of oral syringes for administration or oral, enteral or nebulizer solutions). July 2013.

Guideline

- SCGOPHCG Medicines Management Policy and Procedure
- SCGOPHCG Nursing Practice Guideline Medication Management – Nurse Initiated Medicines
- SCGOPHCG Clinical Practice Guideline Induced Sputum Collection
- SCGOPHCG Clinical Practice Guideline Oxygen Therapy

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| Std 4: Medication Safety



| Std 5: Comprehensive Care



| Std 6:

Communicating for Safety



| Std 8: Recognising and Responding to Acute

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